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Connecting structure for liquid container

Abstract

A connecting structure for a liquid container includes a receiving groove. The liquid container having a suction tube is disposed in the receiving groove. A top end of the receiving groove is engaged with a connecting base. A front side of the connecting base is pivotally connected to a locking assembly protruding frontwards. A liftable base being movable along a top-bottom direction and a front-rear direction is disposed in an inner side of the connecting base, and has a bottom frame engaging with a sleeve that is for being connected to and communicating with a liquid pipeline and fits around the suction tube. Two extending arms extending frontwards out of the bottom frame are pivotally connected to the locking assembly. A left side and a right side of the bottom frame are pivotally connected to two swinging arms that extend backwards and are pivotally connected to the connecting base.

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References Cited

U.S. PATENT DOCUMENTS

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2013/0284735	12/2012	Oesterle	220/23.4	B65D 21/0201

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Background/Summary

BACKGROUND OF THE INVENTION

Technical Field

(1) The present invention relates generally to a structure for installing a liquid container, and more particularly to a connecting structure for a liquid container.

Description of Related Art

(2) When adding a liquid to the conventional liquid dispensers, a bottle containing the liquid is turned upside down to insert into the liquid dispensers, making the liquid flow into the liquid dispensers via a way of gravity. Alternatively, a bottle is engaged with a cap to make the bottle communicate with a suction tube of the cap, and the liquid in the bottle is pumped into the liquid dispensers via a pump for dispensing.

(3) The conventional liquid dispensers have drawbacks. With the way of gravity to make the liquid flow into the liquid dispensers, when the bottle is turned upside down to provide the liquid, the liquid dispensers have to be stable and cannot shake to prevent the bottle that is turned upside down from disengaging from the liquid dispensers to cause leakage. Such a structure is not suitable for applying to a field that the bottle has to move with the liquid dispensers. Moreover, with the structure of the liquid dispensers of which the bottle is engaged with the cap, engaging the bottle with the cap is troubling and time-consuming. Therefore, a structure allowing the liquid dispenser to engage with the bottle rapidly is needed, so as to facilitate the liquid in the bottle to flow into the liquid dispenser.

BRIEF SUMMARY OF THE INVENTION

(4) In view of the above, the primary objective of the present invention is to provide a connecting structure for a liquid container, which could allow a sleeve of a liquid pipeline to be rapidly connected to a liquid container, and a liquid in the liquid container would not leak even if the liquid container is tilted when the liquid container is engaged with the sleeve.

(5) The present invention provides a liquid container connecting structure including a receiving groove, a liquid container, and a locking structure. The liquid container is disposed in the receiving

groove, wherein a top end of the liquid container has a suction tube. The locking structure includes a connecting base, a locking assembly, two swinging arms, and a sleeve, wherein the connecting base is engaged with a top side of the receiving groove. A left side and a right side of the connecting base respectively have a side plate, and a left side and a right side of the locking assembly respectively have a side arm, wherein a rear end of each of the two side arms is pivotally connected to a front side of one of the two side plates. The two side arms extend along a front-rear direction and extend frontwards out of the connecting base, wherein two front ends of the two side arms are connected to a pressing plate. The liftable base is disposed between the two side plates and has a bottom frame. The liftable base moves along a top-bottom direction in the connecting base and moves along the front-rear direction in the connecting base. Two extending arms respectively extend frontwards along the front-rear direction out of a left side and a right side of the bottom frame. A front end of each of the two extending arms is pivotally connected to a front side of each of the two side arms. The left side and the right side of the bottom frame are respectively and pivotally connected to the front end of each of the two swinging arms. A rear end of each of the two swinging arms is pivotally connected to a rear side of one of the two side plates. The sleeve is engaged with the bottom frame and fits around the suction tube.

(6) With the aforementioned design, the sleeve could fit around the suction tube of the liquid container by pressing the locking assembly to drive the liftable base and the sleeve to decline. At that time, even if the liquid container connecting structure is tilted to make the liquid container tilt, the liquid in the liquid container would not leak outwards. Moreover, when the liquid container is needed to be replaced, the liquid container could be released for users to withdraw and replace by flipping the locking assembly upwards to drive the liftable base and the sleeve to lift, thereby the sleeve for connecting to the liquid pipeline could be rapidly connected to or disconnected to the liquid container.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

- (1) The present invention will be best understood by referring to the following detailed description of some illustrative embodiments in conjunction with the accompanying drawings, in which
- (2) FIG. 1 is a perspective view of the liquid container connecting structure according to an embodiment of the present invention;
- (3) FIG. 2 is a partially exploded view of the liquid container connecting structure according to the embodiment of the present invention;
- (4) FIG. 3 is a partially enlarged perspective view of FIG. 1 seen from another direction, showing a part of the left side of the liquid container connecting structure;
- (5) FIG. 4 is a front view of the liquid container connecting structure according to the embodiment of the present invention;
- (6) FIG. 5 is a sectional view along the 5-5 line in FIG. 4;
- (7) FIG. 6 is a sectional view along the 6-6 line in FIG. 4, showing the top portion of the liquid container connecting structure;
- (8) FIG. 7 is an exploded view of the liquid container and the locking structure of the present invention;
- (9) FIG. 8 is an exploded view of the liquid container and the locking structure of the present invention seen from another direction;
- (10) FIG. 9A is a schematic view, showing a section of a part of the locking structure; and
- (11) FIG. 9B is a schematic view, showing the liftable base and the sleeve shown in FIG. 9A are lifted.

DETAILED DESCRIPTION OF THE INVENTION

(12) In the description below, the terms “top”, “bottom”, “front”, “rear”, “left”, “right”, “upper”, “lower”, and their derivatives, should be interpreted from the perspective view of the present invention in FIG. 2. It is also understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to limit the scope of the present invention. A connecting structure **100** for a liquid container according to an embodiment of the present invention is illustrated in FIG. 1 to FIG. 7 and includes a receiving groove **10**, a liquid container **20**, and a locking structure **30**.

(13) The receiving groove **10** could be provided in certain devices. For example, the receiving groove **10** in the current embodiment is provided on, but not limited to, a front surface of a case A1 of a hot spring generator A. The receiving groove **10** has a top end and a bottom end and has a receiving space **12** therein, wherein the receiving space **12** is adapted to receive the liquid container **20**.

(14) As shown in FIG. 2, FIG. 3, and FIG. 6 to FIG. 8, the liquid container **20** is disposed in the receiving space **12** of the receiving groove **10**. A bottom portion of the liquid container **20** abuts against the bottom end of the receiving groove **10**. A top end of the liquid container **20** has a suction tube **243** that is a tube disposed along a top-bottom direction and communicates with an inside of the liquid container **20**. The suction tube **243** could be a plastic tube, or an elastic silicon tube, could bend upon experiencing a force and could restore to an original position after releasing the force.

(15) The locking structure **30** includes a connecting base **32**, a locking assembly **34**, a liftable base **36**, two swinging arms **38**, and a sleeve **31**. The connecting base **32** is engaged with the receiving groove **10** and is located at a top side of the receiving space **12**. A left side and a right side of the connecting base **32** respectively have a side plate **321**, wherein a front side of each of the two side plates **321** has a front axial hole **3211** penetrating along a left-right direction, and a rear side of each of the two side plates **321** has a rear axial hole **3212** penetrating along the left-right direction. A left side and a right side of the locking assembly **34** respectively have a side arm **341**, wherein a rear end of each of the two side arms **341** has a first axial rod **3411**. Each of the first axial rods **3411** pivotally penetrates through each of the front axial holes **3211**, so that the rear end of each of the two side arms **341** is pivotally connected to the front side of each of the two side plates **321**. The two side arms **341** extend along a front-rear direction and extend frontwards out of the connecting base **32**, wherein a front side of each of the two side arms **341** has a front pivot hole **3412** penetrating along the left-right direction, and two front ends of the two side arms **341** are connected to a pressing plate **342**.

(16) The liftable base **36** is disposed between the two side plates **321** of the connecting base **32** and is located directly above the liquid container **20**. The liftable base **36** is restricted in the connecting base **32** to lift or decline along the top-bottom direction or move along the front-rear direction. The liftable base **36** has a bottom frame **361**, wherein two extending arms **3612** respectively extend frontwards along the front-rear direction out of a left side and a right side of a front end of the bottom frame **361**. A front end of each of the two extending arms **3612** has a front pivot rod **3613**, wherein each of the front pivot rods **3613** pivotally penetrates one of the front pivot holes **3412**, so that the front end of each of the two extending arms **3612** is pivotally connected to the front side of each of the two side arms **341**. A left side and a right side of a rear end of the bottom frame **361** respectively have a rear pivot rod **3614**, and a front end of each of the two swinging arms **38** has a rear pivot hole **381**, wherein each of the rear pivot rods **3614** of the bottom frame **361** pivotally penetrates one of the rear pivot holes **381**, so that the left side and the right side of the rear end of the bottom frame **361** are respectively and pivotally connected to the front end of each of the two swinging arms **38**. A rear end of each of the two swinging arms **38** has an inner axial hole **382** that aligns with one of the two rear axial holes **3212**, wherein two second axial rods **383** respectively and pivotally penetrate through the two rear axial holes **3212** and the two inner axial holes **382**, so that the rear end of each of the two swinging arms **38** is pivotally connected to the rear side of one

of the two side plates **321**. Therefore, the two side plates **321** of the connecting base **32**, the two side arms **341** of the locking assembly **34**, the liftable base **36**, and the two swinging arms **38** form a four-bar linkage.

(17) The sleeve **31** is engaged with the bottom frame **361**, wherein a bottom portion of the sleeve **31** detachably fits around the suction tube **243** of the top end of the liquid container **20**. In the current embodiment, when the receiving groove **10** is provided in the case A1 of the hot spring generator A, a top portion of the sleeve **31** is adapted to be connected to and communicate with a connecting tube **33**, and the sleeve **31** is connected to a pipeline in the hot spring generator A via the connecting tube **33**, so that a liquid, such as a hot spring concentrate, in the liquid container **20** could be supplied to the pipeline in the hot spring generator A. Alternatively, in other embodiments, a liquid source inside the case A1 could supply a liquid to the liquid container **20** via the connecting tube **33** and the sleeve **31**.

(18) Referring to FIG. 6, FIG. 9A, and FIG. 9B, when replacing the liquid container **20** in the receiving groove **10** of the liquid container connecting structure **100** of the present invention, a user flips the pressing plate **342** of the locking assembly **34** upwards to make a front side of the locking assembly **34** pivots around the first axial rod **3411**, driving the liftable base **36** and the sleeve **31** to lift along the top-bottom direction, so that the sleeve **31** could be detached from the suction tube **243** of the liquid container **20**, and the liquid container **20** could be withdrawn from the receiving groove **10**. Then a liquid container **20** that is replaced is disposed into the receiving groove **10**, and the pressing plate **342** of the locking assembly **34** is pressed to make the locking assembly **34** pivot back to an original position, driving the liftable base **36** and the sleeve **31** to decline along the top-bottom direction, so that the sleeve **31** could fit around a suction tube **243** of the liquid container **20** again, thereby connecting the liquid container **20** and the sleeve **31** rapidly.

(19) As shown in FIG. 2 and FIG. 6 to FIG. 9A, the top end of the receiving groove **10** has an engaging portion **14** therein, wherein a front surface of the engaging portion **14** has a recess **141**, and a bottom portion of the engaging portion **14** has an opening **142**. A tubing base **16** passes through the opening **142** of the engaging portion **14**, and has a tubing hole **161** therein. The liftable base **36** is located directly below the tubing base **16**. The connecting tube **33** that is connected to the top portion of the sleeve **31** passes upwards through the tubing hole **161** to enter the case A1 to be connected to the pipeline in the hot spring generator A.

(20) A front side of the connecting base **32** has a panel frame **322**, wherein an upper half portion of the panel frame **322** is engaged with and fixed to the recess **141** of the front surface of the engaging portion **14**, and a lower half portion of the panel frame **322** extends downwards along the top-bottom direction out of the recess **141**. A middle of the lower half portion of the panel frame **322** has an opening **3221**. A front-end edge of one of the two side plates **321** is connected to a left side edge of the lower half portion of the panel frame **322**, and a front-end edge of the other side plate **321** is connected to a right side edge of the lower half portion of the panel frame **322**. The two side arms **341** of the locking assembly **34** respectively penetrate through a left side and a right side of the opening **3221** of the panel frame **322**. A middle of the pressing plate **342** has a positioning hole **343**. The bottom frame **361** extends frontwards along the front-rear direction to form an elastic arm **3615**, wherein a middle of the elastic arm **3615** has an elastic structure **3616** that is curved. A front end of the elastic arm **3615** has a positioning member **3617**, wherein the positioning member **3617** is fitted into the positioning hole **343** of the locking assembly **34** for positioning.

(21) A bottom end of the sleeve **31** forms a fitting opening **311** that is bell-shaped, wherein the fitting opening **311** could guide the suction tube **243** to enter the sleeve **31**, which facilitates the sleeve **31** to fit around a top portion of the suction tube **243**, thereby the sleeve **31** could be movably connected to the bottom frame **361** in a float-like way. More specifically, the sleeve **31** could slightly move in a space between the sleeve **31** and the bottom frame **361** along the front-rear direction, thereby absorbing the deviation between the sleeve **31** and the bottom frame **361**. In the current embodiment, a restricting hole **3611** is provided in a middle of the bottom frame **361**. The

sleeve **31** is a plastic tube that is elastic and deformable, and forms a neck **312** between the fitting opening **311** and the top portion of the sleeve **31**. A top edge of the neck **312** of the sleeve **31** forms a first flange **314**, and a bottom edge of the neck **312** of the sleeve **31** forms a second flange **315**, wherein a diameter of the neck **312** is smaller than a diameter of the first flange **314** and a diameter of the second flange **315**. The sleeve **31** is inserted into the restricting hole **3611** from top down, so that the neck **312** of the sleeve **31** passes through the restricting hole **3611** to be fixed on the bottom frame **361** of the liftable base **36**. The diameter of the first flange **314** and the diameter of the second flange **315** are larger than a diameter of the restricting hole **3611**, and the diameter of the neck **312** of the sleeve **31** is smaller than the diameter of the restricting hole **3611**, thereby the sleeve **31** could move in the restricting hole **3611** along the front-rear direction relative to the bottom frame **361**. A periphery of the top portion of the sleeve **31** is engaged with a guiding assembly **313**. A left side and a right side of the guiding assembly **313** respectively have a guiding rod **3131**, wherein the two guiding rods **3131** are both a rod extending upwards along the top-bottom direction, and respectively pass through a left side and a right side of the tubing hole **161** in a slidable way along the top-bottom direction, thereby the guiding assembly **313** could be guided by a tubular wall of the tubing hole **161** to slide along the top-bottom direction.

(22) With the guiding assembly **313** passing through the tubing hole **161**, which allows the sleeve **31** to be guided by the tubular wall of the tubing hole **161** to lift or decline along the top-bottom direction, and with the restricting hole **3611** of the liftable base **36** fitting around a periphery of the neck **312** of the sleeve **31**, the liftable base **36** could be restricted to lift or decline along the top-bottom direction and move along the front-rear direction. As shown in FIG. **8** to FIG. **9A**, a micro switch **35** is disposed on a rear surface of the panel frame **322**, and a front side of the bottom frame **361** forms a protrusion **3618** corresponding to a position of the micro switch **35**. When the liftable base **36** lifts or declines, the protrusion **3618** of the bottom frame **361** could trigger the micro switch **35**, so that the micro switch **35** that is triggered could detect whether the liftable base **36** is located at a high position or a low position. If the liftable base **36** is located at the low position, the micro switch **35** could recognize that the locking structure **30** is in a status of connecting to the liquid container **20**. If the liftable base **36** is located at the high position, the micro switch **35** could recognize that the locking structure **30** is in a status of releasing the liquid container **20**.

(23) As shown in FIG. **8**, the liquid container **20** includes a bottle **22** and a suction tube assembly **24**. The bottle **22** has a body **221** and a neck **222** connected to a top end of the body **221**. The suction tube assembly **24** has a cap **241**, wherein an inner side of the cap **241** has a recessing groove **242**, and a top edge of an outer peripheral surface of the cap **241** has a flange **2411**. The cap **241** is fitted in the neck **222** of the bottle **22**, and the flange **2411** of the cap **241** abuts against a top edge of the neck **222** of the bottle **22** for positioning. The suction tube **243** is engaged with a middle of a bottom portion of the recessing groove **242**, wherein at least a part of the suction tube **243** is located in the recessing groove **242**, and a bottom end of the suction tube **243** communicates with the bottle **22**, so that the suction tube **243** could be detachably engaged with the bottle **22**. In the current embodiment, the suction tube **243** passes through and is engaged with the middle of the bottom portion of the recessing groove **242**. An upper half portion of the suction tube **243** is located in the recessing groove **242**, and a top end of the upper half portion of the suction tube **243** extends to align with a top end of the **241**. A lower half portion of the suction tube **243** extends into the body **221** of the bottle **22**. When the sleeve **31** fits around the suction tube **243**, a bottom side of the sleeve **31** enters the recessing groove **242**. When replacing the liquid container **20** of the connecting structure **100** of the present invention, as the suction tube assembly **24** provided with the suction tube **243** is detachably engaged with the bottle **22**, the suction tube assembly **24** could be engaged with different bottles **22** and is reusable, so that only the bottle **22** is needed to replace when replacing the liquid container **20**, saving the cost of replacing the suction tube assembly **24**.

(24) It must be pointed out that the embodiments described above are only some preferred embodiments of the present invention. All equivalent structures which employ the concepts

disclosed in this specification and the appended claims should fall within the scope of the present invention.

Claims

1. A connecting structure for use with a liquid container, comprising: a receiving groove; the liquid container disposed in the receiving groove, wherein a top end of the liquid container has a suction tube; and a locking structure comprising a connecting base, a locking assembly, two swinging arms, and a sleeve, wherein the connecting base is engaged with a top side of the receiving groove; a left side and a right side of the connecting base respectively have a side plate, and a left side and a right side of the locking assembly respectively have a side arm, wherein a rear end of each of the two side arms is pivotally connected to a front side of one of the two side plates; the two side arms extend along a front-rear direction and extend frontwards out of the connecting base, wherein two front ends of the two side arms are connected to a pressing plate; a liftable base is disposed between the two side plates and has a bottom frame; the liftable base moves along a top-bottom direction in the connecting base and moves along the front-rear direction in the connecting base; two extending arms respectively extend frontwards along the front-rear direction out of a left side and a right side of the bottom frame; a front end of each of the two extending arms is pivotally connected to a front side of each of the two side arms; the left side and the right side of the bottom frame are respectively and pivotally connected to the front end of each of the two swinging arms; a rear end of each of the two swinging arms is pivotally connected to a rear side of one of the two side plates; the sleeve is engaged with the bottom frame and fits around the suction tube.
2. The connecting structure for use with the liquid container as claimed in claim 1, wherein the liquid container comprises a bottle and a suction tube assembly; a top portion of the bottle has a neck; the suction tube assembly has a cap; an inner side of the cap has a recessing groove, and a top edge of an outer peripheral surface of the cap has a flange; the cap is fitted in the neck of the bottle, and the flange of the cap abuts against at a top edge of the neck of the bottle for positioning; the suction tube is engaged with a middle of a bottom portion of the recessing groove, wherein at least a part of the suction tube is located in the recessing groove.
3. The connecting structure for use with the liquid container as claimed in claim 1, wherein a top end of the receiving groove has an engaging portion therein, and a tubing hole is disposed on a bottom portion of the engaging portion; a periphery of a top portion of the sleeve is engaged with a guiding assembly; the guiding assembly passes through the tubing hole in a slidable way along the top-bottom direction; the sleeve is movably connected to the bottom frame and is configured to slightly move within a space between the sleeve and the bottom frame along the front-rear direction to absorb the deviation between the sleeve and the bottom frame.
4. The connecting structure for use with the liquid container as claimed in claim 1, wherein a top end of the receiving groove has an engaging portion therein; a bottom portion of the engaging portion has an opening; a tubing base passes through the opening of the engaging portion and has a tubing hole therein; the liftable base is located directly below the tubing base; a connecting tube is connected to a top portion of the sleeve and passes through the tubing hole.
5. The connecting structure for use with the liquid container as claimed in claim 4, wherein a periphery of a top portion of the sleeve is engaged with a guiding assembly; a left side and a right side of the guiding assembly respectively have a guiding rod; the two guiding rods are both a rod extending upwards along the top-bottom direction, and respectively pass through a left side and a right side of the tubing hole in a slidable way along the top-bottom direction.
6. The connecting structure for use with the liquid container as claimed in claim 5, wherein a middle of the bottom frame has a restricting hole; the sleeve is a tube that is elastic and forms a neck; a top edge of the neck of the sleeve forms a first flange, and a bottom edge of the neck of the sleeve forms a second flange; the neck of the sleeve passes through the restricting hole to be fixed

on the bottom frame of the liftable base; a diameter of the first flange and a diameter of the second flange are larger than a diameter of the restricting hole; a diameter of the neck is smaller than the diameter of the restricting hole.

7. The connecting structure for use with the liquid container as claimed in claim 4, wherein a front side of the connecting base has a panel frame; an upper half portion of the panel frame is engaged with and fixed to a front surface of the engaging portion, and a middle of a lower half portion of the panel frame has an opening; the two side arms of the locking assembly respectively penetrate through a left side and a right side of the opening of the panel frame; a front-end edge of each of the two side plates is connected to the lower half portion of the panel frame.

8. The connecting structure for use with the liquid container as claimed in claim 7, wherein a micro switch is disposed on a rear surface of the panel frame, and a front side of the bottom frame forms a protrusion corresponding to a position of the micro switch; when the liftable base lifts or declines, the protrusion of the bottom frame triggers the micro switch.

9. The connecting structure for use with the liquid container as claimed in claim 1, wherein a bottom end of the sleeve forms a fitting opening that is bell-shaped.

10. The connecting structure for use with the liquid container as claimed in claim 1, wherein a middle of the pressing plate has a positioning hole; the bottom frame extends frontwards along the front-rear direction to form an elastic arm; a middle of the of the elastic arm has an elastic structure that is curved; a front end of the elastic arm has a positioning member fitting into the positioning hole.
